	n mod 3	(n+2) mod 3	(n+4) mod 3	(n+6) mod 3
$n = 0 \bmod 3$] $n=0$	$0 \to 0 \bmod 3$	$2 \to 0+2 \bmod 3$	$1 \to 0+4 \bmod 3$	$0 \to 0+6 \bmod 3$
$n = 1 \bmod 3$] $n=1$	$1 \to 1 \bmod 3$	$0 \to 1+2 \bmod 3$	$2 \to 4+1 \bmod 3$	$1 \to 1+6 \bmod 3$
$n = 2 \bmod 3$] $n=2$	$2 \to 2 \bmod 3$	$1 \to 2+2 \bmod 3$	$0 \to 2+4 \bmod 3$	$2 \to 2+6 \bmod 3$
Think of modular arithmetic $a-b=$] →	$n = \bmod 3$	$n+2 = \bmod 3$	$n+4 = \bmod 3$	$n+6 = \bmod 3$

Modular Arithmetic:

HW4 DATA250:

Q20: Given $a \equiv b \bmod m$, where $m \geq 2$ and $c > 0$,

Show that $ac \equiv bc \bmod mc$

Proof: Since $a \equiv b \pmod m$, we have $m \mid (a-b)$, so $a-b = Km$ for some integer K.

Multiply both sides by c: $ac - bc = K(mc)$

This means $mc \mid (ac - bc)$, so $ac \equiv bc \pmod{mc}$ ✓

Q20: What is $\gcd(x, y)$

20a if $x = 2^2 3^3 5^5$ and $y = 2^5 3^3 5^2$

$\gcd(x,y) = 2^{\min(2,5)} \cdot 3^{\min(3,3)} \cdot 5^{\min(5,2)} = 2^2 \cdot 3^3 \cdot 5^2 = 4 \cdot 27 \cdot 25 = \boxed{2700}$

20b if $x = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 9 \cdot 11 \cdot 13$ and $y = 2^{11} \cdot 3^9 \cdot 17^{14}$

$\gcd(x,y)$ $2^{\min(1,11)} \cdot 3^{\min(1,9)} \cdot 5^{\min(1,0)} \cdot 7^{\min(1,0)} \cdot 9^{\min(1,0)} \cdot 11^{\min(1,0)} \cdot 13^{\min(1,0)} \cdot 17^{\min(0,14)}$

$2^1 \cdot 3^1 \cdot 5^0 \cdot 7^0 \cdot 9^0 \cdot 11^0 \cdot 13^0 \cdot 17^0$

$= 2 \cdot 3 = \boxed{6}$

Every composite in these 4 consecutive odd numbers is divisible by 3, so use mod 3.

Not prime

9 exists so things work work.

Question X: 3, 5, 7, 9, 11 (9 is composite)

✓ No such four consecutive odd primes exist.

5, 7, 9, 11 (9 is composite)

7, 9, 11, 13 (9 is composite)

9, 11, 13, 15 (9, 15 composite)

11, 13, 15, 19 (15 composite)

3 divisible by 3 ~~but also prime~~

mod 3

n

n+2

n+4

n+6

For any four consecutive odd numbers one must be divisible by 3:

If $n \equiv 0 \pmod 3$, then n divisible by 3

If $n \equiv 1 \pmod 3$, then $n+2 \equiv 0 \bmod 3$

If $n \equiv 2 \pmod 3$, then $n+4 \equiv 0 \bmod 3$

Since $n > 3$, the one divisible by 3 will be composire (>3).